# Linux `tree` Command Notes

The `tree` **command** is used to display directory structures in a **tree-like format**, making it easier to visualize files and folders compared to `ls -R`.

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## 🔹 1. Basic Usage

```
tree
```

**Example Output:**

```
.
├── Documents
│   ├── file1.txt
│   └── file2.txt
├── Downloads
│   └── setup.exe
└── notes.txt

2 directories, 3 files
```

👉Shows the hierarchy of the current directory.

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## 🔹 2. Show Hidden Files

```
tree -a
```

**Example Output:**

```
.
├── .hiddenfile
├── Documents
│   └── file.txt
└── notes.txt
```

👉Includes hidden files (starting with `.`).

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## 🔹 3. Limit Directory Depth

```
tree -L 2
```

**Example Output:**

```
.
├── Documents
│   ├── file1.txt
│   └── Notes
├── Downloads
└── notes.txt
```

👉Only shows up to 2 levels deep.

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## 🔹 4. Show File Sizes

```
tree -s
```

**Example Output:**

```
[1024] file1.txt
[2048] file2.txt
```

👉Displays file sizes in bytes.

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## 🔹 5. Human-Readable Sizes

```
tree -h
```

**Example Output:**

```
[1.0K] file1.txt
[2.0K] file2.txt
```

👉Sizes shown in KB, MB, GB.

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## 🔹 6. Show Full Path of Files

```
tree -f
```

**Example Output:**

```
./Documents/file1.txt
./Documents/file2.txt
./Downloads/setup.exe
./notes.txt
```

👉Prints full paths for each file.

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## 🔹 7. Show Directories Only

```
tree -d
```

**Example Output:**

```
.
├── Documents
├── Downloads
└── Notes
```

👉Only lists directories.

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## 🔹 8. Sort by Modification Time

```
tree -t
```

👉Lists files and directories sorted by modification time.

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## 🔹 Installation

`tree` might not be installed by default.

```
sudo apt install tree     # Debian/Ubuntu
sudo yum install tree     # RHEL/CentOS
brew install tree         # macOS (Homebrew)
```

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## 🔹 Comparison with `ls -R`

- `ls -R` → Recursive listing but flat, less readable.

- `tree` → Visual hierarchy, easy to understand.

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## 🔹 Summary of Common Options

| Option | Meaning | Example |
|---|---|---|
| `-a` | Show hidden files | `tree -a` |
| `-L n` | Limit directory depth to *n* | `tree -L 2` |
| `-s` | Show file sizes in bytes | `tree -s` |
| `-h` | Human-readable file sizes | `tree -h` |
| `-f` | Show full path for each file | `tree -f` |
| `-d` | Show directories only | `tree -d` |
| `-t` | Sort by last modification time | `tree -t` |

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👉The `tree` **command** is a handy visual tool to explore directory structures in the terminal.